

This is CS50

CS50's Introduction to Computer Science

OpenCourseWare

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Substitution

Problem to Solve

In a substitution cipher, we “encrypt” (i.e., conceal in a reversible way) a message by replacing every letter with another letter. To do so, we use a *key*: in this case, a mapping of each of the letters of the alphabet to the letter it should correspond to when we encrypt it. To “decrypt” the message, the receiver of the message would need to know the key, so that they can reverse the process: translating the encrypt text (generally called *ciphertext*) back into the original message (generally called *plaintext*).

A key, for example, might be the string `NQXPOMAFTRHLZGECYJIUWSKDVB`. This 26-character key means that `A` (the first letter of the alphabet) should be converted into `N` (the first character of the key), `B` (the second letter of the alphabet) should be converted into `Q` (the second character of the key), and so forth.

A message like `HELLO`, then, would be encrypted as `FOLLE`, replacing each of the letters according to the mapping determined by the key.

In a file called `substitution.c` in a folder called `substitution`, create a program that enables you to encrypt messages using a substitution cipher. At the time the user executes the program, they should decide, by providing a command-line argument, on what the key should be in the secret message they'll provide at runtime.

Demo

```
$ ./substitution
Usage: ./substitution key
$
```

Recorded with **asciinema**

Specification

Design and implement a program, `substitution`, that encrypts messages using a substitution cipher.

- Implement your program in a file called `substitution.c` in a directory called `substitution`.
- Your program must accept a single command-line argument, the key to use for the substitution. The key itself should be case-insensitive, so whether any character in the key is uppercase or lowercase should not affect the behavior of your program.
- If your program is executed without any command-line arguments or with more than one command-line argument, your program should print an error message of your choice (with `printf`) and return from `main` a value of `1` (which tends to signify an error) immediately.

- If the key is invalid (as by not containing 26 characters, containing any character that is not an alphabetic character, or not containing each letter exactly once), your program should print an error message of your choice (with `printf`) and return from `main` a value of `1` immediately.
- Your program must output `plaintext:` (without a newline) and then prompt the user for a `string` of plaintext (using `get_string`).
- Your program must output `ciphertext:` (without a newline) followed by the plaintext's corresponding ciphertext, with each alphabetical character in the plaintext substituted for the corresponding character in the ciphertext; non-alphabetical characters should be outputted unchanged.
- Your program must preserve case: capitalized letters must remain capitalized letters; lowercase letters must remain lowercase letters.
- After outputting ciphertext, you should print a newline. Your program should then exit by returning `0` from `main`.

You might find one or more functions declared in `ctype.h` to be helpful, per manual.cs50.io (<https://manual.cs50.io/>).

Walkthrough



How to Test

Correctness

In your terminal, execute the below to check your work's correctness.

```
check50 cs50/problems/2025/x/substitution
```

► How to Use **debug50**

Style

Execute the below to evaluate the style of your code using `style50`.

```
style50 substitution.c
```

How to Submit

In your terminal, execute the below to submit your work.

```
submit50 cs50/problems/2025/x/substitution
```

